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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Hanoi, July 20, 2026

To: The Admissions Committee
University of Michigan-Dearborn
Dearborn, MI 48128, USA

Subject: Letter of Recommendation for Xuan Thanh Trinh

Dear Members of the Admissions Committee,

I am Dr. Tran Thanh Son, a Lecturer in the School of Electrical and Electronic Engineering at Hanoi University of Science and Technology (HUST). Having earned my PhD from the Shibaura Institute of Technology in Japan, I am a core member of the Power Grid and Renewable Energy Lab at HUST, where I lead a research group focusing on power system optimization, the integration of renewable energy sources, and the operation of distribution grids. I write to you today to recommend Xuan Thanh Trinh for admission to the PhD program at the University of Michigan-Dearborn.

Over the past year, I directly supervised Thanh in his role as my primary research assistant to develop and co-author a journal publication. Throughout this collaboration, I have been impressed by his ability to bridge the gap between abstract theoretical models and rigorous computational execution. In practice, I often design intricate optimization frameworks, but translating these mathematical constraints into robust code is a significant challenge, which Thanh completely resolved by taking full ownership of this computational backend. Rather than simply running simulations, he proactively verified his outputs against physical laws to ensure the core methodology remained structurally sound. This high level of technical reliability allowed me to focus my efforts entirely on advanced algorithm design, making him an invaluable asset to our lab's productivity.

I particularly appreciated Thanh's ability to work independently and adapt methodically to shifting project requirements. When the computational models encountered roadblocks, he did not just report the errors to me. Instead, he took the initiative to diagnose the root causes and typically prepared several potential solutions to address their theoretical implications. This approach allowed us to resolve issues efficiently while staying aligned with the project's overarching goals. Additionally, whenever I introduced new methodological constraints, Thanh adapted thoughtfully. Rather than immediately discarding his previous work, he researched the new tools and carefully compared their performance against our existing methods before making a complete transition. This combination of independent problem-solving and methodical adaptability indicates that he is well-prepared for the demands of a doctoral program.

Beyond his technical skills, Thanh excels at communicating complex research with absolute clarity. Internally, he translated raw simulation data into clear visual trends, enabling fast, data-driven decisions. Externally, he structured our manuscripts to preempt reviewer critiques and confidently presented our findings to academic audiences. This rare ability to both execute and persuasively articulate advanced research establishes him as a truly exceptional co-author.

Xuan Thanh Trinh is a rare talent who combines mathematical rigor, computational execution speed, and profound academic maturity. I give him my highest recommendation for the PhD program at the University of Michigan-Dearborn, confident that he will be an immediate force multiplier for your research team. Please feel free to contact me if you require any further information.

Sincerely,

Dr. Tran Thanh Son
Lecturer, School of Electrical and Electronic Engineering
Hanoi University of Science and Technology (HUST), Vietnam
Email: son.tranthanh@hust.edu.vn